Questions

Q1.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

(i) Solve, for $-\frac{\pi}{2} < x < \frac{\pi}{2}$

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$$\tan^2\left(2x + \frac{\pi}{4}\right) = 3$$

(5)

(ii) Solve, for $0 < \theta < 360°$

$$(2\sin\theta - \cos\theta)^2 = 1$$

giving your answers, as appropriate, to one decimal place.

(5)

(Total for question = 10 marks)

(**Q10 WMA12/01, Oct 2021**)

Q2.

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In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

Solve, for $-180° < \theta \leq 180°$, the equation

$$3\tan(\theta + 43°) = 2\cos(\theta + 43°)$$

(Total for question = 6 marks)

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(**Q05 WMA12/01, June 2022**)

Q3.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

(a) Solve, for $0 < \theta \leq 360°$ the equation

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$$2 \tan \theta + 3 \sin \theta = 0$$

giving your answers, as appropriate, to one decimal place.

(5)

(b) Hence, or otherwise, find the smallest positive solution of

$$2 \tan (2x + 40°) + 3 \sin (2x + 40°) = 0$$

giving your answer to one decimal place.

(2)

(Total for question = 7 marks)

(**Q03 WMA12/01, Oct 2023**)

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Q4.

(i) (a) Find, in ascending powers of x, the 2nd, 3rd and 5th terms of the binomial expansion of

$$(3 + 2x)^6$$

(3)

For a particular value of x, these three terms form consecutive terms in a geometric series.

(b) Find this value of x.

(3)

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(ii) In a **different** geometric series,

- the first term is $\sin^2\theta$
- the common ratio is $2\cos\theta$
- the sum to infinity is $\frac{8}{5}$

(a) Show that

$$5\cos^2\theta - 16\cos\theta + 3 = 0$$

(3)

(b) Hence find the exact value of the 2nd term in the series.

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(3)

(Total for question = 12 marks)

(Q10 WMA12/01, Oct 2023)

Q5.

(i) Solve, for $0 \leq \theta < \pi$, the equation

$$\sin 3\theta - \sqrt{3}\cos 3\theta = 0$$

giving your answers in terms of π

(3)

(ii) Given that

$$4\sin^2 x + \cos x = 4 - k, \quad 0 \leq k \leq 3$$

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(a) find $\cos x$ in terms of k

(3)

(b) When $k = 3$, find the values of x in the range $0 \leq x < 360°$

(3)

(Total for question = 9 marks)

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(Q09 WMA12/01, Specimen papers)

Q6.

(a) Show that the equation

$$\cos\theta - 1 = 4\sin\theta\tan\theta$$

can be written in the form

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$$5\cos^2\theta - \cos\theta - 4 = 0$$

(4)

(b) Hence solve, for $0 \leqslant x < \frac{\pi}{2}$

$$\cos 2x - 1 = 4\sin 2x\tan 2x$$

giving your answers, where appropriate, to 2 decimal places.

(4)

(Total for question = 8 marks)

(Q09 WMA12/01, June 2019)

Q7.

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(a) Show that the equation

$$8\tan\theta = 3\cos\theta$$

may be rewritten in the form

$$3\sin^2\theta + 8\sin\theta - 3 = 0$$

(3)

(b) Hence solve, for $0 \leq x \leq 90°$, the equation

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$$8 \tan 2x = 3 \cos 2x$$

giving your answers to 2 decimal places.

(4)

(Total for question = 7 marks)

(Q07 WMA12/01, Jan 2020)

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Q8.

(i) Show that

$$\tan\theta + \frac{1}{\tan\theta} \equiv \frac{1}{\sin\theta\cos\theta} \qquad \theta \neq \frac{n\pi}{2} \qquad n \in \mathbb{Z}$$

(3)

(ii) Solve, for $0 \leq x < 90°$, the equation

$$3\cos^2(2x + 10°) = 1$$

giving your answers in degrees to one decimal place.

(Solutions based entirely on graphical or numerical methods are not acceptable.)

(4)

(Total for question = 7 marks)

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(Q07 WMA12/01, Oct 2020)

Q9.

(a) Show that the equation

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$\frac{3\sin\theta\cos\theta}{2\sin\theta - 1} = 5\tan\theta \qquad \sin\theta \neq \frac{1}{2}$

can be written in the form

$$3\sin^3\theta + 10\sin^2\theta - 8\sin\theta = 0$$

(4)

(b) Hence solve, for $-\frac{\pi}{4} < x < \frac{\pi}{4}$

$$\frac{3\sin 2x\cos 2x}{2\sin 2x - 1} = 5\tan 2x$$

giving your answers to 3 decimal places where appropriate.

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(4)

(Total for question = 8 marks)

(**Q06 WMA12/01, Jan 2021**)

Q10.

In this question you must show all stages of your working.

Solutions based entirely on calculator technology are not acceptable.

(i) Solve, for $-\frac{\pi}{2} < x < \pi$ the equation

$$5\sin(3x + 0.1) + 2 = 0$$

giving your answers, **in radians**, to 2 decimal places.

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(4)

(ii) Solve, for $0 < \theta < 360°$, the equation

$$2\tan\theta\sin\theta = 5 + \cos\theta$$

giving your answers, **in degrees**, to one decimal place.

(5)

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(Total for question = 9 marks)

(Q08 WMA12/01, Jan 2023)

Q11.

$$f(x) = kx^3 - 15x^2 - 32x - 12 \qquad \text{where } k \text{ is a constant}$$

Given $(x - 3)$ is a factor of $f(x)$,

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(a) show that $k = 9$

(2)

(b) Using algebra and showing each step of your working, fully factorise $f(x)$.

(4)

(c) Solve, for $0 \leq \theta < 360°$, the equation

$$9\cos^3\theta - 15\cos^2\theta - 32\cos\theta - 12 = 0$$

giving your answers to one decimal place.

(2)

(Total for question = 8 marks)

(Q06 WMA12/01, June 2019)

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Q12.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

(i) Solve, for $0 < \theta < 360°$, the equation

$$3\sin(\theta + 30°) = 7\cos(\theta + 30°)$$

giving your answers to one decimal place.

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(4)

(ii) (a) Show that the equation

$$3 \sin^3 x = 5 \sin x - 7 \sin x \cos x$$

can be written in the form

$$\sin x (a \cos^2 x + b \cos x + c) = 0$$

where a, b and c are constants to be found.

(b) Hence solve for $-\frac{\pi}{2} \leqslant x \leqslant \frac{\pi}{2}$ the equation

$$3 \sin^3 x = 5 \sin x - 7 \sin x \cos x$$

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(6)

(Total for question = 10 marks)

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(Q08 WMA12/01, June 2021)

Q13.

Solutions based entirely on graphical or numerical methods are not acceptable in this question.

(i) Solve, for $0 \leq \theta < 180°$, the equation

$$3 \sin (2\theta - 10°) = 1$$

giving your answers to one decimal place.

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(4)

(ii) The first three terms of an arithmetic sequence are

$$\sin\alpha, \quad \frac{1}{\tan\alpha} \quad \text{and} \quad 2\sin\alpha$$

where α is a constant.

(a) Show that $2 \cos \alpha = 3 \sin^2 \alpha$

(3)

Given that $\pi < \alpha < 2\pi$,

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(b) find, showing all working, the value of α to 3 decimal places.

(5)

(Total for question = 12 marks)

(Q09 WMA12/01, Oct 2019)

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Q14.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

(a) Show that

$$3\cos\theta(\tan\theta\sin\theta + 3) = 11 - 5\cos\theta$$

may be written as

$$3\cos^2\theta - 14\cos\theta + 8 = 0$$

(3)

(b) Hence solve, for $0 < x < 360°$

$$3\cos 2x(\tan 2x\sin 2x + 3) = 11 - 5\cos 2x$$

giving your answers to one decimal place.

(4)

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(Total for question = 7 marks)

(Q09 WMA12/01, June 2023)

Q15.

In this question you must show detailed reasoning.

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Solutions relying entirely on calculator technology are not acceptable.

(i) Solve, for $0 \leq x < 360°$, the equation

$$\sin x \tan x = 5$$

giving your answers to one decimal place.

(6)

(ii)

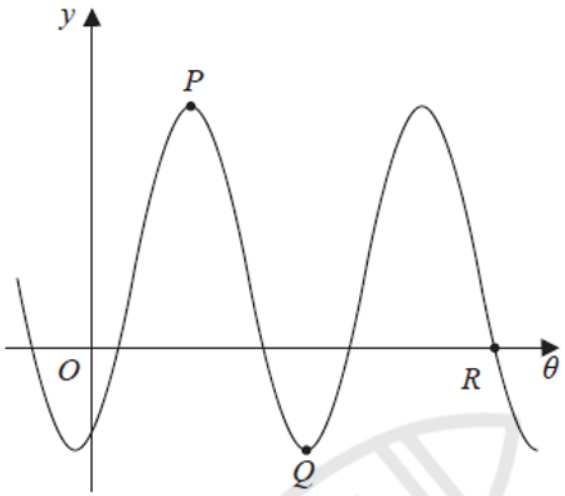


Figure 1

Figure 1 shows a sketch of part of the curve with equation

$$y = A\sin\left(2\theta - \frac{3\pi}{8}\right) + 2$$

where A is a constant and θ is measured in radians.

The points P, Q and R lie on the curve and are shown in Figure 1.

Given that the y coordinate of P is 7

(a) state the value of A,

(1)

(b) find the exact coordinates of Q,

(3)

(c) find the value of θ at R, giving your answer to 3 significant figures.

(4)

(Total for question = 14 marks)

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(**Q09 WMA12/01, Jan 2024**)

Q16.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

(i) Solve, for $-90° < x < 90°$, the equation

$$3 \sin (2x - 15°) = \cos (2x - 15°)$$

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giving your answers to one decimal place.

(4)

(ii) Solve, for $0 < \theta < 2\pi$, the equation

$$4 \sin^2 \theta + 8 \cos \theta = 3$$

giving your answers to 3 significant figures.

(4)

(Total for question = 8 marks)

(**Q07 WMA12/01, Jan 2022**)

Q17.

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In this question you must show detailed reasoning.

Solutions relying entirely on calculator technology are not acceptable.

(a) Show that the equation

$$(3\cos\theta - \tan\theta)\cos\theta = 2$$

can be written as

$$3\sin^2\theta + \sin\theta - 1 = 0$$

(3)

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(b) Hence solve for $-\frac{\pi}{2} \leqslant x \leqslant \frac{\pi}{2}$

$$(3\cos 2x - \tan 2x)\cos 2x = 2$$

(5)

(Total for question = 8 marks)

(**Q05 WMA12/01, Oct 2022**)

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Q18.

$$f(x) = 6x^3 + 17x^2 + 4x - 12$$

(a) Use the factor theorem to show that $(2x + 3)$ is a factor of $f(x)$.

(2)

(b) Hence, using algebra, write $f(x)$ as a product of three linear factors.

(4)

(c) Solve, for $\frac{\pi}{2} < \theta < \pi$, the equation

$$6\tan^3\theta + 17\tan^2\theta + 4\tan\theta - 12 = 0$$

giving your answers to 3 significant figures.

(2)

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(Total for question = 8 marks)

(**Q03 WMA12/01, Jan 2020**)

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